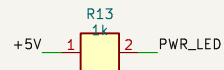
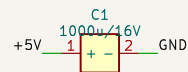
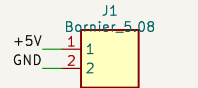
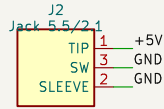
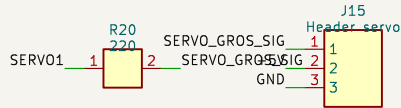


A PROTEGER EN AMONT (fusible/disjoncteur) : charges inductives : snubber cote charge.

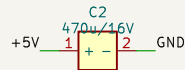
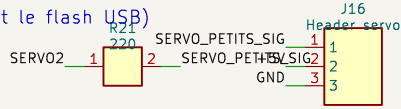
ALIMENTATION 5V (3A recommandé) – jack OU bornier.  
D5 protège le rail si l'USB du DevKit est branché en même temps.



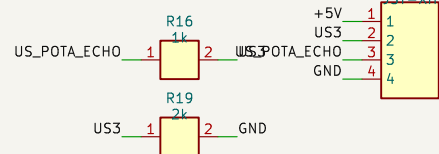
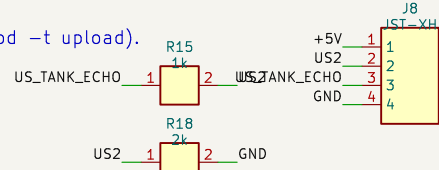
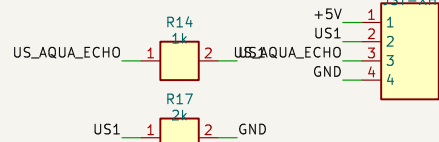
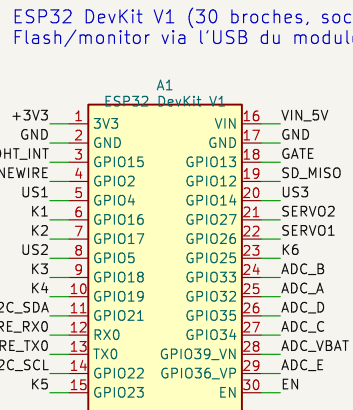
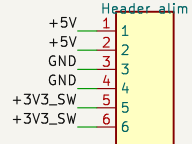
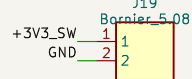
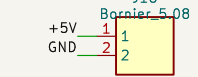
SERVOS NOURRISSEURS 5V  
GPIO12 (strapping MTDI) : pull-down R22, signal série R20.



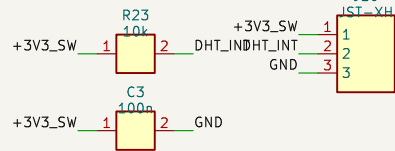
TOUS les GPIO libres sur header (Dupont),  
y compris RX0/TX0 (les laisser libres pendant le flash USB)  
et GPIO36/39 (entrees seules).



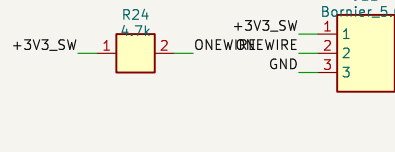
Distribution 5V / 3V3-capteurs / GND :  
borniers a vis + rail header.



AIR : DHT11 (option DHT22, -DUSE\_DHT22) - pull-up 10k.



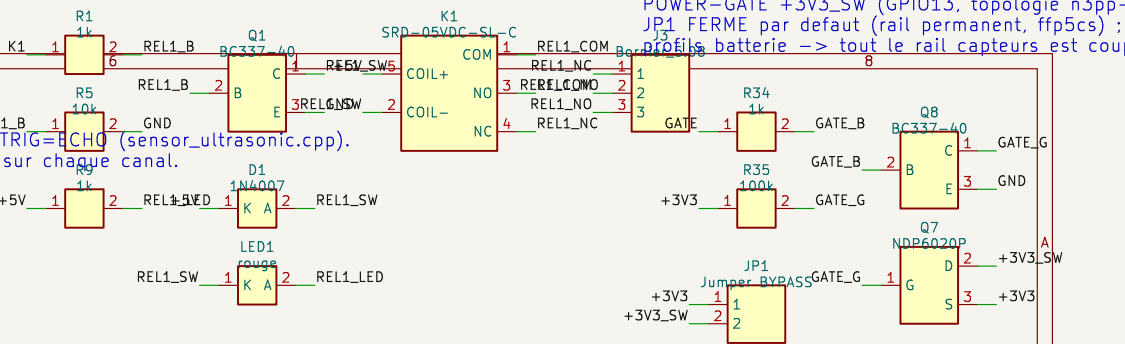
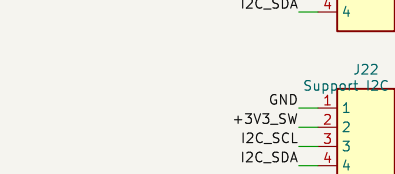
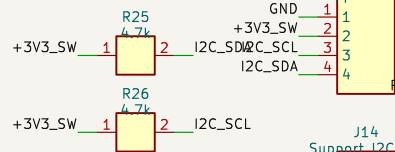
EAU : DS18B20 (1-Wire, pull-up 4.7k).



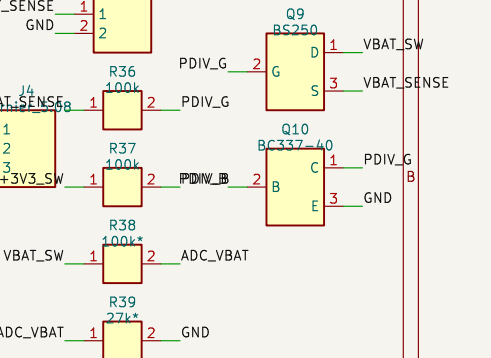
LUMIERE : LDR déportée + 10k (GPIO34 = entrée seule).



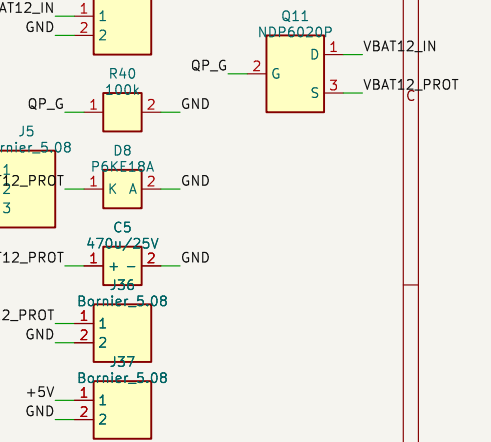
I2C : OLED SSD1306 0x3C + header extension (DS3231 si besoin)



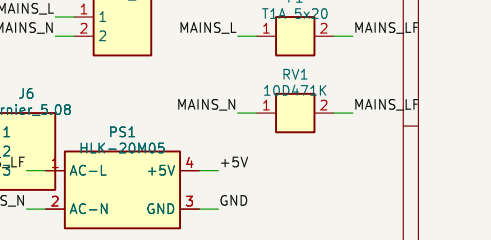
PONT DIVISEUR VBAT COMMUTE : actif seulement  
R38/R39 SELON PROFIL : 1S Li-ion = 100k/100



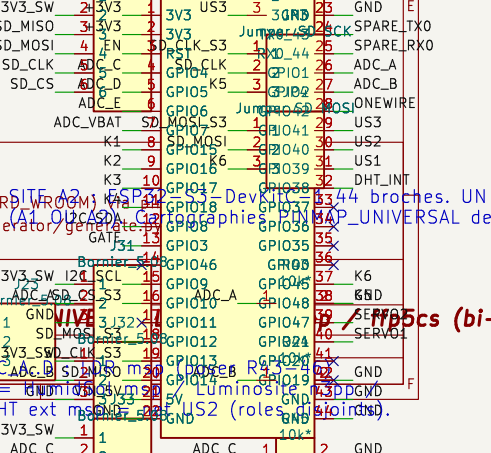
PROFIL BUS 12V SOLAIRE : fusible lame 7,5-10A  
anti-inversion P-MOSFET, TVS 18V, réservoir ; b  
(module MP1584/XL4015 faible Iq sur entretoise



PROFIL SECTEUR : Hi-Link HLK-20M05 EMBARQUÉ  
+ varistance 300VAC. Le corps du module enjambe  
secteur/logique (fente fraisée dessous). ZONE 2.



microSD : module SPI 3V3 sockete. JP2/3/4 : s  
CS/SCK/MOSI = 63 stat1 (1-2, default) ou WR0  
MISO = 11 stat2 (1-2, default) ou WR0  
MISO = 11 stat2 (1-2, default) ou WR0

[illegible]